PRO MINDS

MATH TUTORING · NAPERVILLE

ACT Math Check-Up

12 questions. Every topic the ACT math section tests. Work them on paper, check your answers with the worked key inside, and look at *which topics* the misses came from — that pattern tells you exactly what to study. Every question below is tagged with its topic and with the class in our ACT Math Prep course that teaches it.

Print it, sit somewhere quiet, and try each one honestly — a wrong answer here is useful information, not a grade.

The Check-Up

1. If $3(x - 4) + 5 = 2x + 7$, what is the value of x? **Topic: Linear equations**

A. 10 **B.** 12

C. 14 **D.** 16

2. A jacket that originally cost $85 is on sale for 20% off. What is the sale price? **Topic: Percents**

A. $65 **B.** $68

C. $72 **D.** $102

3. A line passes through the points (2, 3) and (6, 11). What is the slope of the line? **Topic: Slope of a line**

A. 2 **B.** 1/2

C. 5/2 **D.** 4

4. If $x + y = 10$ and $x - y = 4$, what is the value of x? **Topic: Systems of equations**

A. 3 **B.** 4

C. 6 **D.** 7

5. If $f(x) = x^2 - 3x + 2$, what is the value of $f(-2)$? **Topic: Functions**

A. 0 **B.** 12

C. 4 **D.** 20

6. Which expression is equivalent to $(2x^3)(3x^2)$? **Topic: Exponent rules**

A. $6x^5$
B. $5x^5$
C. $6x^6$
D. $36x^5$

7. What is the larger solution of the equation $x^2 - 5x + 6 = 0$? **Topic: Quadratics**

A. 1
B. 2
C. 3
D. 6

8. The two shorter sides of a right triangle measure 6 inches and 8 inches. What is the length of the longest side? **Topic: Right triangles**

A. 9 inches
B. 10 inches
C. 12 inches
D. 14 inches

9. A circle has a radius of 5 centimeters. What is the area of the circle? **Topic: Circles**

A. 10π cm^2
B. 20π cm^2
C. 25π cm^2
D. 50π cm^2

10. In a right triangle, angle A is not the right angle. The side opposite angle A has length 3, and the hypotenuse has length 5. What is sin(A)? **Topic: Trigonometry**

A. 3/5
B. 4/5
C. 3/4
D. 5/4

11. A student's four quiz scores are 72, 85, 90, and 81. What is the average (mean) of the scores? **Topic: Averages**

A. 80
B. 81
C. 82
D. 84

12. A bag holds 4 red marbles and 6 blue marbles. One marble is chosen without looking. What is the probability that it is red? **Topic: Probability**

A. 1/5
B. 2/5
C. 3/5
D. 2/3

Worked Answer Key

Don't just check right or wrong — read the step where yours went different. Each answer shows the full working and the exact spot where students usually slip.

1. Answer: C Linear equations · Class 3

Distribute: 3x − 12 + 5 = 2x + 7, so 3x − 7 = 2x + 7. Subtract 2x from both sides: x − 7 = 7, so x = 14.

Where students get stuck: Forgetting to distribute the 3 to BOTH terms inside the parentheses — 3(x − 4) is 3x − 12, not 3x − 4.

2. Answer: B Percents · Class 1

20% off means paying 80% of the price: 0.80 × $85 = $68.

Where students get stuck: Finding 20% of 85 ($17) and stopping — that is the discount, not the price. Either subtract it ($85 − $17 = $68) or multiply by 80% directly.

3. Answer: A Slope of a line · Class 7

Slope = (change in y) ÷ (change in x) = (11 − 3) ÷ (6 − 2) = 8 ÷ 4 = 2.

Where students get stuck: Dividing the wrong way (run over rise) and getting 1/2. Slope is rise over run: y-change on top.

4. Answer: D Systems of equations · Class 4

Add the two equations: (x + y) + (x − y) = 10 + 4, so 2x = 14 and x = 7. (Then y = 3.)

Where students get stuck: Solving for the wrong letter, or stopping halfway. Underline what the question asks for before you start.

5. Answer: B Functions · Class 5

Replace every x with −2: $(-2)^2 - 3(-2) + 2 = 4 + 6 + 2 = 12$.

Where students get stuck: Sign errors: $(-2)^2$ is +4, not −4, and −3 × (−2) is +6, not −6. Use parentheses around the negative number every time.

6. Answer: A Exponent rules · Class 2

Multiply the numbers: 2 × 3 = 6. Add the exponents on the same base: $x^3 \cdot x^2 = x^5$. Result: $6x^5$.

Where students get stuck: Multiplying the exponents (x^6) instead of adding them. Exponents add when you multiply powers of the same base.

7. Answer: C Quadratics · Class 4

Factor: two numbers that multiply to 6 and add to −5 are −2 and −3, so $(x - 2)(x - 3) = 0$. The solutions are $x = 2$ and $x = 3$; the larger is 3.

Where students get stuck: Answering 2 because it is the first solution found. When a question asks for the larger (or smaller) value, finish both and then choose.

8. Answer: B Right triangles · Class 6

$a^2 + b^2 = c^2$, so $c^2 = 6^2 + 8^2 = 36 + 64 = 100$, and $c = 10$ inches. (This is the classic 6-8-10 triangle — worth recognizing on sight.)

Where students get stuck: Adding 6 + 8 = 14. The sides must be squared first, then added, then square-rooted.

9. Answer: C Circles · Class 6

Area = $\pi r^2 = \pi \times 5^2 = 25\pi$ cm².

Where students get stuck: Mixing up area (πr^2) with circumference ($2\pi r = 10\pi$ here). If the question asks for area, the radius gets squared.

10. Answer: A Trigonometry · Class 7

sin = opposite ÷ hypotenuse = 3 ÷ 5 = 3/5.

Where students get stuck: Using the adjacent side (4/5 is cos(A)). Fix the ratio words first: SOH — Sine = Opposite over Hypotenuse.

11. Answer: C Averages · Class 8

Add: 72 + 85 + 90 + 81 = 328. Divide by the number of scores: 328 ÷ 4 = 82.

Where students get stuck: Dividing by the wrong count, or averaging only the first and last numbers. Mean = total ÷ how many values there are.

12. Answer: B Probability · Class 8

Probability = (favorable outcomes) ÷ (total outcomes) = 4 ÷ 10 = 2/5.

Where students get stuck: Using only the blue count or the red-to-blue ratio (4/6 = 2/3). The bottom number is ALL the marbles, red and blue together.

Formula & Basics Sheet

The must-know facts, organized by topic, with what each one is for. Keep this next to you while you practice.

Percents (Class 1). Part = percent × whole. Percent change = change ÷ original × 100. "20% off" means pay 80%.

Exponents & roots (Class 2). x^a · x^b = x^(a+b) · (x^a)^b = x^(ab) · $x^0 = 1$ · x^(−a) = 1/x^a · √x = x^(1/2). Multiply powers: add exponents. Raise a power to a power: multiply them.

Linear equations & inequalities (Class 3). Distribute to BOTH terms: 3(x − 4) = 3x − 12. Whatever you do to one side, do to the other. For inequalities, flip the sign when you multiply or divide by a negative.

Systems, quadratics & polynomials (Class 4). Systems: add or subtract the equations to cancel a letter, or substitute. Quadratics: try factoring first — (x − r)(x − s) = 0 gives x = r and x = s. Backup: x = (−b ± √(b² − 4ac)) ÷ 2a.

Functions & sequences (Class 5). f(a) means: replace every x with a — use parentheses for negatives. Arithmetic sequence: a_n = a_1 + (n − 1)d. Geometric: a_n = a_1 · r^(n−1).

Geometry (Class 6). Triangle area = ½ × base × height. Right triangles: a² + b² = c² (c is the longest side). Circle: area = πr², circumference = 2πr. Angles in a triangle add to 180°; a straight line is 180°.

Coordinate geometry & trig (Class 7). Slope = $(y_2 - y_1) \div (x_2 - x_1)$. Line equation: y = mx + b. Midpoint: average the x's, average the y's. Distance: $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. Trig: sin = opposite/hypotenuse, cos = adjacent/hypotenuse, tan = opposite/adjacent (SOH-CAH-TOA).

Statistics, probability & counting (Class 8). Mean = total ÷ count. Median = the middle value when ordered. Probability = favorable outcomes ÷ ALL outcomes. Counting: multiply the number of choices at each step.

What your misses are telling you

Match each missed question to its row. Two or more misses in the same topic is not a knowledge problem — it is one fixable habit, and it is exactly what the class works on, from the basics, step by step.

Missed	Topic	Taught in
Q1	Linear equations	Class 3
Q2	Percents	Class 1

Q3	Slope of a line	Class 7
Q4	Systems of equations	Class 4
Q5	Functions	Class 5
Q6	Exponent rules	Class 2
Q7	Quadratics	Class 4
Q8	Right triangles	Class 6
Q9	Circles	Class 6
Q10	Trigonometry	Class 7
Q11	Averages	Class 8
Q12	Probability	Class 8

The next step: ACT Math Prep

Eight live online classes built for the **December 12 ACT** — Nov 30, Dec 1, 2, 3, 4, 7, 8 & 9. **Only 6 students. $300 for all 8 classes.** Every topic taught from the basics, step by step, with time to ask questions in every class.

Step 1: Register for the December 12 ACT — late registration runs to **November 29**.
Step 2: Text **ACT** to **312-956-3846** to save a seat.

Not sure yet? Text **FALL** to the same number for a free 15-minute consultation — bring this Check-Up and we will go through the misses together.

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